



PUSL3190 Computing Project

Project Proposal

Vehicle Attendance Verification System

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Chapter 01: Problem Statement

Colombo Dockyard PLC, a major ship repair facility processing over 200 vessels annually, currently relies on a UHF RFID based vehicle attendance system to administer fuel allowances for its workforce of over 1,000 employees. This system is critical for verifying which employees have brought their personal vehicles to work, which is the prerequisite for receiving the monthly fuel allowance. However, the current system is failing to provide accurate verification due to a combination of environmental interference and behavioral exploitation.

The primary technical failure stems from severe electromagnetic interference caused by the Colombo Harbor Authority's separate RFID system, which operates on identical frequency bands in the adjacent port area. This cross-talk manifests as "ID collision" and "dual-card ambiguity." In ID collision scenarios, RFID readers at the Dockyard gate inadvertently detect Harbor Authority access cards, creating false positive logs. In dual-card ambiguity scenarios, employees carrying both cards (for access to both zones) generate conflicting signals, preventing the system from reliably identifying the correct credential. These technical errors have significantly eroded trust in the system's accuracy.

This technical unreliability has enabled widespread behavioral exploitation. The fundamental architectural flaw of the current RFID implementation is that it verifies the presence of an identification card rather than the presence of the vehicle itself. Employees have identified this gap and frequently scan their RFID cards while entering the facility as passengers in other vehicles, on public transport, or as pedestrians. Because the system frequently generates false positives due to interference, these fraudulent manual scans are often indistinguishable from legitimate system errors.

The financial and operational impact of this problem is substantial. Internal audits conducted by the Finance department reveal a consistent 15-20% discrepancy between the number of vehicle entries recorded by the RFID system and the actual physical occupancy of the vehicle parking lots. This discrepancy translates to an estimated monthly financial loss of 225,000 to 300,000 LKR in unjustified fuel allowance payments. Annually, this amounts to approximately 2.7 to 3.6 million LKR in operational leakage.

Furthermore, the problem creates a significant administrative burden. The Finance department allocates approximately 20 hours per month to investigating disputed allowance claims, while the HR department spends considerable time managing employee complaints regarding denied payments. Security personnel are also affected; attempting to mitigate fraud through manual verification during peak shift changes where over 200 vehicles arrive within a 30-minute window causes severe traffic bottlenecks. The manual process requires 30-45 seconds per vehicle, which

is operationally unfeasible during rush hours, forcing security staff to abandon verification to maintain traffic flow. This inconsistency in enforcement further encourages non-compliance.

In summary, the problem is a compounded failure of verification technology that creates financial loss, administrative waste, and operational inefficiency. A solution is required that can verify the physical presence of the vehicle independent of the RFID interference and without relying on manual intervention.

Chapter 02: Project Description

The proposed solution is the development of an automated "Vehicle Attendance Verification System with Multi-Gate Spatial Verification." This system aims to replace the reliance on the compromised RFID infrastructure with an optical verification method that is immune to radio-frequency interference. The core of the solution is a dual-camera License Plate Recognition (LPR) system designed specifically for the challenging conditions of the Colombo Dockyard maritime environment.

System Architecture and Functionality:

The system utilizes computer vision to detect and recognize vehicle license plates as they enter the facility. Two high-definition cameras (Front and Rear) will be deployed at the entry gate to capture vehicle images. This dual-camera setup addresses the issue of occlusion and ensures coverage for vehicles with only rear plates (such as certain older models or motorcycles).

The video feed will be processed locally using the YOLOv8 (You Only Look Once) object detection algorithm. YOLOv8 was selected for its balance of high accuracy and real-time processing speed. Once a license plate is detected, the image region is passed to a Character Recognition neural network based on the CRNN (Convolutional Recurrent Neural Network) architecture with CTC (Connectionist Temporal Classification) loss. This recognition engine will be specifically trained on Sri Lankan license plate formats, including standard provincial codes (e.g., WP, SP), the legacy "Sri" series, and newer format variations.

Following character recognition, the detected plate number is validated against an internal employee-vehicle registration database. This process uses a two-stage matching logic:

1. **Exact Matching:** The system searches for a direct string match in the database.
2. **Fuzzy Matching:** If no exact match is found, the system calculates the Levenshtein distance to identify registered plates that are within a one-character variation (e.g., recognizing "CAB-1234" as "CAB-1234" despite a minor OCR error).

Verification Logic and Spatial Analysis:

The system implements a multi-layer verification approach. The primary layer confirms vehicle presence via optical recognition. A secondary "Spatial Verification" layer is introduced to detect complex fraud. By monitoring entry and exit events across multiple gates (e.g., Gate 1 and Gate 3), the system calculates the travel time between points. If a vehicle appears at two distant gates within a time window that is physically impossible for travel (e.g., 2 minutes for a 1.2km distance), the system flags a "Spatial Impossibility" alert, indicating potential plate cloning or database error.

User Interface and Feedback:

The system provides real-time feedback to drivers and security personnel. An LED display board at the gate will show the verification status (e.g., "CONFIRMED" or "ACCESS DENIED"). For security staff, a manual control interface allows for "ACCEPT" or "REJECT" overrides in cases of uncertain matches, ensuring human-in-the-loop oversight without causing delays. Comprehensive audit logs record every transaction, including the timestamp, recognized plate image, and verification decision, providing the Finance department with indisputable evidence for allowance payments.

Project Objectives:

1. **Develop a Robust Optical Detection System:** Implement a dual-camera setup using YOLOv8 optimized with TensorRT to achieve $\geq 85\%$ detection accuracy in variable maritime lighting conditions.
2. **Implement Region-Specific Character Recognition:** Train a CRNN model specifically for Sri Lankan license plate formats, capable of handling non-standard fonts and worn plates with $\geq 90\%$ recognition precision.
3. **Create a High-Performance Validation Database:** Build a local PostgreSQL database with fuzzy matching capabilities (Levenshtein distance) to validate plates against employee records in under 50ms.
4. **Establish Multi-Gate Spatial Verification:** Develop logic to identify fraudulent activity by analyzing travel times between different facility gates, flagging impossible transit scenarios.
5. **Deploy a Transparent Audit Interface:** Create a secure logging system that records timestamped images and verification results, providing 100% auditability for financial processing.

Project Keywords:

1. Computer Vision
2. License Plate Recognition (LPR/ALPR)
3. YOLOv8 Object Detection
4. Spatial Verification
5. Industrial IoT

Chapter 03: Research Gap

While Automatic License Plate Recognition (ALPR) is a mature technology globally, significant research gaps exist regarding its application in specific industrial contexts and local environments.

1. **Mitigation of RFID Interference in Industrial Zones:** Existing literature extensively covers RFID systems and ALPR systems separately. However, there is a lack of research on using optical ALPR specifically as a counter-measure for radio-frequency interference in congested industrial zones like harbors. Most studies focus on ALPR for traffic management or tolling, not as a replacement for compromised access control systems in high-interference environments where RFID reliability is degraded by cross-talk from neighboring facilities.
2. **Application to Sri Lankan License Plate Formats:** The majority of open-source ALPR datasets and pre-trained models (e.g., OpenALPR) are optimized for European, American, or Chinese license plate formats. Sri Lankan plates present unique challenges, including diverse generations of formats (e.g., "Sri" series vs. modern provincial codes), varying aspect ratios, and non-standard fonts used on older vehicles. There is limited published work on training deep learning models (specifically YOLOv8/CRNN architectures) on Sri Lankan datasets, particularly those degraded by maritime environmental factors like salt spray corrosion and welding dust.
3. **Spatial-Temporal Fraud Detection in Private Facilities:** Current vehicle access control research typically focuses on single-point verification (entry/exit). The concept of using "spatial impossibility" checks—validating the physical feasibility of travel times between multiple internal gates to detect plate cloning or pass-back fraud—is under-researched in the context of private industrial facility management. This project addresses this gap by integrating spatial logic into the verification pipeline.
4. **Low-Cost Edge Deployment:** Many commercial ALPR solutions rely on cloud processing or expensive, specialized hardware. This project explores the feasibility of deploying modern, high-accuracy deep learning models on consumer-grade hardware (standard GPUs) using optimization techniques like TensorRT INT8 quantization, filling a gap in affordable, on-premise industrial deployment strategies.

Chapter 04: Requirements Analysis

Hardware Requirements: The development and deployment of this system require specific hardware components. A computer with a dedicated NVIDIA GPU (RTX 3060 or higher) is necessary for training the deep learning models and running real-time inference. Two IP cameras with 2K resolution and Wide Dynamic Range (WDR) capabilities are required to handle the high-contrast lighting conditions at the dockyard gates. An LED matrix display will be needed for the user interface.

Software and Knowledge Requirements: The project relies heavily on Python programming and deep learning frameworks. Proficiency in PyTorch is required for model development. Knowledge of computer vision libraries such as OpenCV is essential for image preprocessing. Understanding of the YOLO (You Only Look Once) architecture for object detection and CRNN (Convolutional Recurrent Neural Network) for sequence recognition is critical.

Additionally, the project requires knowledge of database management using PostgreSQL to handle employee records and audit logs. Techniques for model optimization, specifically NVIDIA TensorRT, must be learned to ensure the system runs with low latency on the available hardware. Familiarity with "fuzzy matching" algorithms (Levenshtein distance) is also necessary for the database validation logic.

Special Devices/Knowledge: No proprietary industrial controllers are required. However, knowledge of standard TCP/IP networking is needed to interface with the IP cameras. The project also requires an understanding of basic electronics to interface the software with the LED display board via serial or GPIO connections.

Chapter 05: Finance

Budget Plan (Estimated)

The following budget outlines the estimated costs for the hardware and resources required to build the prototype and MVP. Note that these costs are for the project development and will not be reimbursed.

Item Description	Quantity	Estimated Unit Cost (LKR)	Total Cost (LKR)
IP Cameras (2K, WDR, IR)	2	25,000	50,000
LED Display Panel (P10 Outdoor)	1	15,000	15,000
Mounting Brackets & Cables (CAT6)	1 Set	10,000	10,000
Processing Unit (Existing Laptop)	1	0	0
Miscellaneous (Casings, Connectors)	-	10,000	10,000
Total Estimated Cost			85,000

Chapter 06: External Organizations

External Party: Colombo Dockyard PLC is the primary external organization involved. The company is providing the test site (Gate 3), access to the existing vehicle database structure, and permission to install cameras for data collection. They are the ultimate client for this system.

Other Staff / Supporting Staff:

- **Mr. Madusanka Mithrananda (Supervisor):** Provides academic guidance and reviews project milestones.
- **Dockyard Security Personnel:** Will assist in the physical testing phase and provide feedback on the user interface.
- **Dockyard Finance Department Staff:** Will be consulted to define the requirements for the audit logs and export formats.

Chapter 07: Time Frame / Timeline

The project will be executed over a period of approximately 6 months, from October 2025 to April 2026.

Stage	Week													
	2	4	6	8	10	12	14	16	18	20	22	24	26	28
1														
2														
3														
4														
5														
6														

Phase	Duration	Start Date	End Date	Total Cost (LKR)
1. Analysis & Planning	4 Weeks	01 Oct 2025	31 Oct 2025	Problem definition, requirement gathering, gap analysis, proposal submission.
2. Data Collection	3 Weeks	01 Nov 2025	21 Nov 2025	Installation of cameras, collection of 500+ site-

				specific images, data annotation.
3. Model Development	5 Weeks	22 Nov 2025	31 Dec 2025	Training YOLOv8 detection model and CRNN recognition model, optimization.
4. System Integration	4 Weeks	01 Jan 2026	31 Jan 2026	Integration of cameras, AI models, and database; development of UI logic.
5. Testing & Validation	4 Weeks	01 Feb 2026	28 Feb 2026	On-site testing at Gate 3, accuracy verification, latency optimization.
6. Documentation	4 Weeks	01 Mar 2026	31 Mar 2026	Writing final thesis, user manuals, and preparing presentation materials.
7. Final Submission			April 2026	Submission of final report and software artifact.

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Project Initiation Document

Vehicle Attendance Verification System

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Chapter 01: Introduction

Background and Context

Colombo Dockyard PLC is a major ship repair facility that processes more than 200 vessels annually (Colombo Dockyard PLC, 2025). The workforce consists of over 1,000 employees (Colombo Dockyard PLC, 2025). Currently, the facility operates a vehicle attendance verification system based on UHF RFID technology. This system functions to administer fuel allowances for employees. However, the system faces significant issues due to electromagnetic interference from the Colombo Harbor Authority RFID system. This interference leads to technical errors and allows behavioral exploitation by the employees.

This represents the third generation of vehicle attendance systems at Colombo Dockyard. The first system was an Electronic Toll Collection (ETC) solution. It was discontinued because of high dependency on the vendor. The in-house staff lacked the expertise for routine maintenance, which caused long downtimes. The second system was a Hikvision computer vision solution. It failed to meet requirements because of incomplete vehicle coverage, specifically for motorcycles. It also had high false detection rates and an inability to read Sri Lankan license plates in the maritime environment, which contains salt spray and variable lighting.

Subsequently, Colombo Dockyard deployed the UHF RFID system. It functions for employee identity verification and fuel allowance distribution. However, the Harbor Authority uses a UHF RFID system on the same frequency bands. This creates "ID collision," where Harbor cards are mistaken for Dockyard cards, and "dual-card ambiguity," where the system cannot determine which card is detected if an employee carries both.

Problem Statement

The fundamental problem is that the current RFID system verifies the presence of the card but not the presence of the vehicle. This architectural gap allows employees to claim fuel allowances without bringing vehicles to the facility. Due to RF interference creating false allowance credits, the system credibility has reduced. Consequently, employees scan RFID cards while entering as passengers or pedestrians to claim the allowance.

Internal audits indicate a 15-20% discrepancy between RFID-recorded vehicle entries and actual parking lot occupancy (Colombo Dockyard PLC, 2025). This results in an estimated monthly loss of 225,000-300,000 LKR (Colombo Dockyard PLC, 2025). The annual loss is approximately 2.7-3.6 million LKR (Colombo Dockyard PLC, 2025). The Finance department allocates significant

time to investigate disputed claims without definitive evidence. Security personnel face difficulties during peak shift changes. Approximately 200 vehicles arrive within a 30-minute window (Colombo Dockyard PLC, 2025). Manual verification causes delays, while lack of verification allows fraud.

Specific operational metrics include:

- Manual verification requires 30-45 seconds per vehicle (Colombo Dockyard PLC, 2025).
- Security personnel cannot manually verify all entries during peak hours.
- Finance audit standards require verifiable evidence, which RFID does not provide.
- Policy enforcement remains inconsistent.

Scope and Limitations

Included in Project Scope:

The project involves the development of a dual-camera license plate recognition system. It utilizes YOLOv8 for object detection and CRNN for character recognition. The system is specifically designed for Sri Lankan license plate formats. The detected plates will be validated against an employee-vehicle registration database using exact matching and fuzzy matching (Levenshtein distance ≤ 1) for damaged plates (Colombo Dockyard PLC, 2025).

A comprehensive audit logging system will be implemented to record verification decisions with timestamps, plate numbers, employee identities, and match types. The user interface includes an LED display for driver confirmation and manual ACCEPT/REJECT controls for security personnel. The initial deployment is planned for Gate 3.

The system architecture supports three verification layers:

1. Primary optical vehicle presence detection via license plates.
2. Optional RFID fusion when interference is low (temporal correlation within ± 5 seconds) (Colombo Dockyard PLC, 2025).
3. Spatial-temporal impossibility detection across multiple gates to identify physically impossible travel scenarios.

All processing is performed locally on the facility premises. No external cloud communication or data transmission is included.

Excluded from Scope:

Real-time RFID integration is deferred to Phase 2. The current version is designed for standalone ALPR operation. The system does not perform automated fuel allowance calculation; it provides audit logs for the Finance department to calculate payments manually. Mobile application interfaces are excluded. Cloud synchronization is not included to ensure data sovereignty. Motorcycle-specific detection optimization is planned for Phase 2.

Expected Impact and Stakeholders

Primary Stakeholders:

The **Finance Department** will gain the ability to validate fuel allowance claims against actual vehicle attendance. Currently, reconciliation of RFID scans against legitimate entries is not possible. The new system provides exportable audit logs in JSON and CSV formats with timestamp-linked plate images.

Security Personnel will be relieved from manual verification duties during peak hours. Currently, verifying 200+ vehicles in a 30-minute window is not feasible (Colombo Dockyard PLC, 2025). The LED display provides immediate visual confirmation. System-generated audit trails protect security staff from disputes by providing evidence of decisions.

HR and Compliance Management staff handle employee disputes regarding fuel allowance denials. The system provides audit evidence with timestamp-linked photographs and confidence scores, which reduces dispute resolution time.

Secondary Stakeholders:

Legitimate Vehicle Users currently face delays during manual verification. The automated system provides fair processing with sub-1.5 second processing time (Colombo Dockyard PLC, 2025).

Colombo Dockyard Leadership faces decisions regarding technology investments. This project demonstrates an approach using open-source frameworks to avoid vendor lock-in and addresses site-specific environmental challenges.

Relevance and Timeliness

This project addresses the immediate operational issue of system credibility erosion. As Colombo Dockyard expands, the facility projects 300-500 daily entries during peak periods (Colombo Dockyard PLC, 2025). Manual verification cannot scale to this level.

The research is relevant due to the increasing deployment of RFID systems in Sri Lankan industrial zones. When multiple systems coexist, interference occurs. This project demonstrates optical-spectrum alternatives. It also addresses the gap in Sri Lankan license plate recognition research, as most existing studies focus on Western plate formats.

Chapter 02: Business Case

2.1 Business Need

The current system suffers from technical failure (RF interference) and behavioral exploitation. This has eroded confidence in the system accuracy.

The quantified business needs are:

Financial Leakage: Internal audits reveal 15-20% discrepancies in vehicle entries (Colombo Dockyard PLC, 2025). This represents 225,000-300,000 LKR monthly in unverified claims (Colombo Dockyard PLC, 2025).

Administrative Burden: Finance staff spend approximately 20 hours per month investigating disputes. HR staff spend 15 hours per month handling employee complaints. This totals 35+ hours of monthly administrative overhead.

Operational Bottlenecks: Security personnel face unmanageable workloads during shift changes. 200+ vehicles arrive in 30 minutes (Colombo Dockyard PLC, 2025). Manual verification of 30-45 seconds per vehicle (Colombo Dockyard PLC, 2025) creates queues.

Audit Compliance Risk: Audit standards require evidence for payments. The RFID system confirms card presence, not vehicle presence. This creates a compliance vulnerability.

System Credibility: Frequent false-positive credits cause employees to lose trust in the system.

2.2 Business Objectives

The project aims to achieve the following objectives:

Objective 1 - Fraud Reduction: Reduce unverified claims from 15-20% to below 2% within six months (Colombo Dockyard PLC, 2025). Validated through parking lot occupancy counts.

Objective 2 - Financial Recovery: Achieve monthly savings of 180,000-240,000 LKR (Colombo Dockyard PLC, 2025). The target is to recover the investment cost within 2-3 months.

Objective 3 - Administrative Efficiency: Reduce Finance and HR administrative burden from 35+ hours to less than 5 hours per month.

Objective 4 - Operational Throughput: Maintain peak-hour processing capacity. Target processing time is ≤ 1.5 seconds per vehicle (Colombo Dockyard PLC, 2025).

Objective 5 - Audit Compliance: Achieve 100% audit log integrity with comprehensive verification trails.

Objective 6 - Scalability: Establish infrastructure to support 300-500 daily vehicle entries without increasing security staff (Colombo Dockyard PLC, 2025).

Chapter 03: Project Objectives

The specific technical objectives are:

Objective 1 - Dual-Camera Detection: Deploy a synchronized front and rear camera system. Technical specifications include 2K resolution and Wide Dynamic Range (WDR) to handle maritime lighting.

Objective 2 - Recognition Engine: Implement CRNN+CTC character recognition achieving $\geq 90\%$ precision on Sri Lankan plates (Colombo Dockyard PLC, 2025). The system must handle five canonical formats and automatically insert missing dashes (e.g., AB1234 to AB-1234).

Objective 3 - Database Validation: Deploy an employee-vehicle database with exact and fuzzy matching (Levenshtein distance ≤ 1) (Colombo Dockyard PLC, 2025). Query time must be ≤ 50 ms (Colombo Dockyard PLC, 2025).

Objective 4 - Audit Infrastructure: Generate tamper-evident audit logs. Fields include timestamp, plate number, match type, and decision. Daily backup with SHA-256 integrity validation is required.

Objective 5 - User Interface: Deploy an LED display with manual ACCEPT/REJECT controls. The target is $\geq 90\%$ user acceptance from security personnel (Colombo Dockyard PLC, 2025).

Objective 6 - Performance: Achieve end-to-end processing latency ≤ 1.5 seconds (95th percentile) (Colombo Dockyard PLC, 2025). Memory usage is to be maintained ≤ 4 GB through TensorRT optimization (Colombo Dockyard PLC, 2025).

Chapter 04: Literature Review

Introduction

The literature review examines Automatic License Plate Recognition (ALPR), RFID limitations, and industrial computer vision. Databases such as IEEE Xplore and Google Scholar were searched for publications between 2016-2025.

ALPR Algorithms

Anagnostopoulos et al. (2008) survey license plate recognition, identifying plate localization and character segmentation as fundamental steps. Variable lighting is noted as a key challenge.

Du et al. (2013) review deep learning in ALPR. They emphasize that accuracy depends on training data diversity. Generic datasets do not generalize well to new environments. This confirms the necessity of collecting site-specific data at Colombo Dockyard.

Laroca et al. (2018) demonstrate that YOLO detectors achieve real-time performance. They report inference times below 50ms on GPU hardware. This supports the selection of YOLOv8 for this project.

Li et al. (2019) propose end-to-end detection using CNNs and RNNs. This integrated approach outperforms segmented methods. The project architecture follows this YOLO + CRNN combination.

Henry et al. (2020) address multinational plate recognition using generalized sequence detection. This is relevant to the diverse plate formats found in Sri Lanka.

Björklund et al. (2019) demonstrate that automated license plate recognition using CNNs trained on synthetic data can achieve competitive performance. However, they note that synthetic data struggles to replicate specific environmental degradation, reinforcing the need for site-specific data collection for maritime conditions.

Character Recognition

Shi et al. (2017) introduce CRNN with CTC loss. This architecture treats text recognition as a sequence problem, which is robust to irregular character spacing common in license plates.

Silva and Gunawardana (2021) specifically address ALPR for Sri Lankan vehicles. They identify challenges such as provincial code prefixes and legacy "Sri" series plates. Their work validates the need for a local dataset.

Industrial Deployment

NVIDIA Corporation (2024) outlines TensorRT optimization techniques. FP16 quantization allows real-time inference on consumer GPUs. This is essential for achieving the 1.5-second latency target.

Tabrizi and Cavus (2016) discuss hybrid recognition models. This supports the project's strategy of using both exact and fuzzy matching.

Research Gaps

The review identifies the following gaps:

1. Lack of research on replacing RFID with ALPR specifically to mitigate interference.
2. Spatial impossibility detection across multiple gates is unexplored in ALPR context.
3. Generic ALPR datasets do not cover Sri Lankan plates in maritime industrial conditions.

Chapter 05: Method of Approach

Research Design

The project follows an experimental prototype methodology. Stage 0 involves empirical gap analysis to quantify the fraud rate. Subsequent stages include POC development, MVP implementation, and testing.

Data Collection

A dataset of 500-1,000 images will be collected from Gate 3 cameras. The collection will cover diverse conditions including rain and variable lighting. Images will be manually annotated with bounding boxes and character transcriptions to create ground truth. The dataset will include representation of cars, vans, and motorcycles. Privacy protocols will be followed, ensuring driver faces are masked in training data.

Algorithms and Technical Stack

Detection Pipeline:

- Model: YOLOv8-Small architecture.
- Optimization: TensorRT (INT8/FP16) for inference speedup.
- Logic: Confidence threshold ≥ 0.7 is applied to reduce false positives.

Recognition Pipeline:

- Model: CRNN with CTC loss.
- Post-processing: Format validation logic is applied. For example, inserting dashes into sequences like "AB1234" to match the "AB-1234" database format.

Database Matching:

- Database: MS SQL Server.
- Matching: Exact matching via SQL and fuzzy matching via Levenshtein distance calculation in Python.

Development Environment:

- Language: Python 3.9+.

- Libraries: PyTorch, OpenCV.
- Hardware: Laptop with NVIDIA RTX 3060 GPU.

Evaluation Metrics

Detection Performance:

- Precision: Target $\geq 90\%$ (Colombo Dockyard PLC, 2025).
- Recall: Target $\geq 85\%$ (Colombo Dockyard PLC, 2025).

System Performance:

- Latency: Target $p95 \leq 1.5$ seconds from vehicle entry to confirmation (Colombo Dockyard PLC, 2025).

Business Impact:

- Fraud Reduction: Target is to reduce discrepancies from 15-20% to $< 2\%$ (Colombo Dockyard PLC, 2025).

Project Management

A hybrid Agile-Waterfall approach is adopted. The timeline consists of fixed stages, but development within stages follows two-week sprints.

Chapter 06: Conceptual Diagram

Diagram 1: High-Level System Architecture

The following diagram illustrates the data flow from the cameras to the database and the user interface.

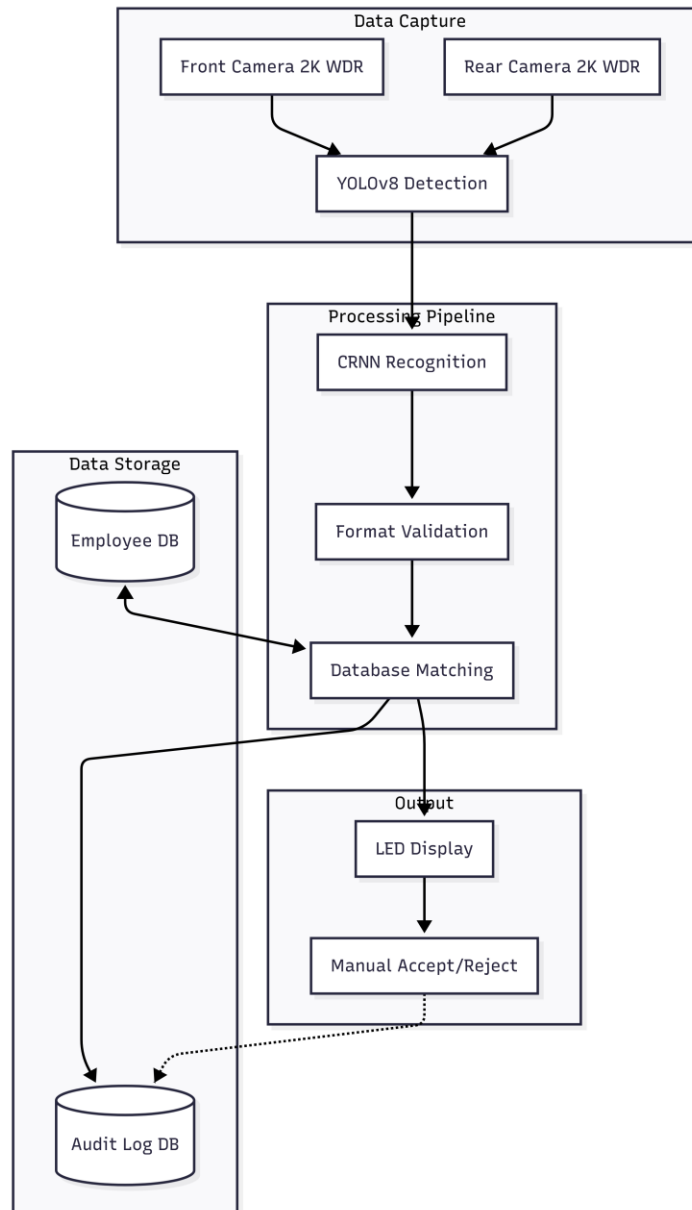
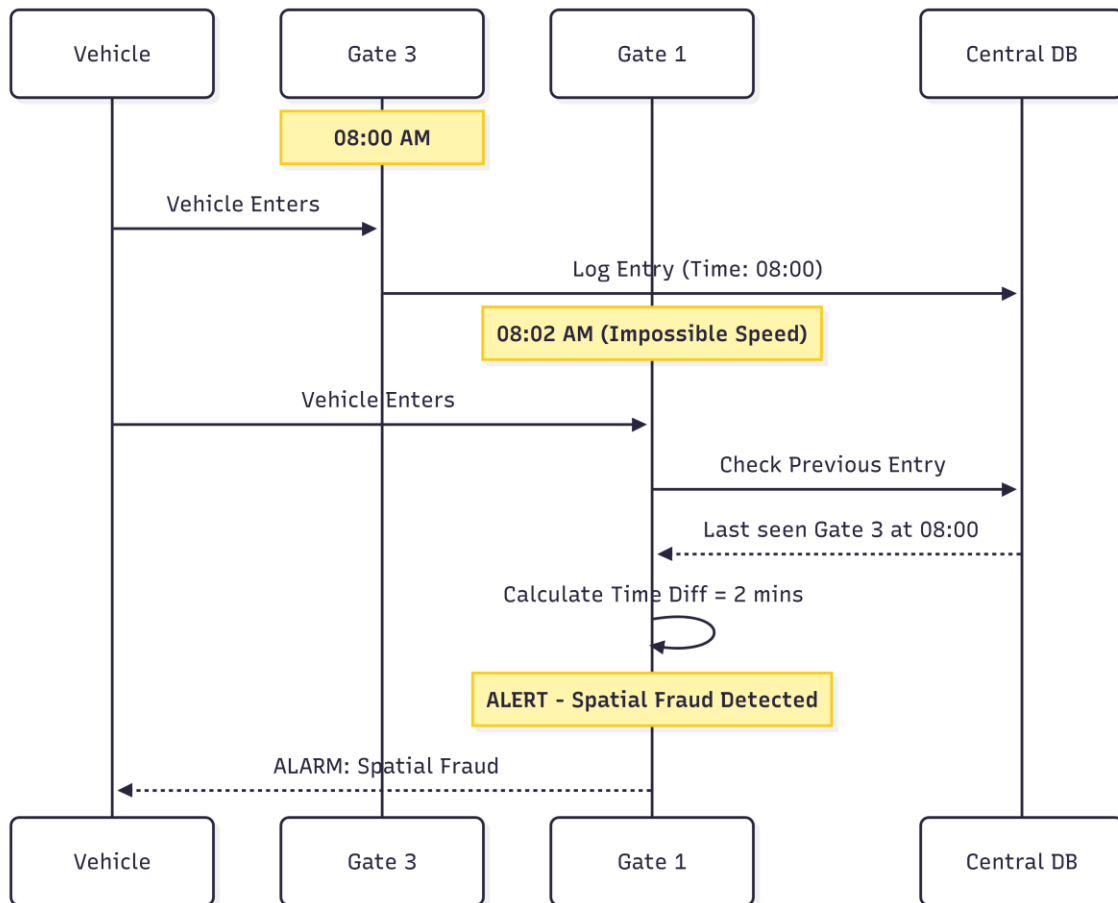


Diagram 2: Multi-Gate Spatial Verification Logic

This diagram demonstrates the logic used to detect physically impossible travel times between gates.



Chapter 07: Initial Project Plan

Project Timeline (October 2025 - April 2026)

Stage	Week													
	2	4	6	8	10	12	14	16	18	20	22	24	26	28
Stage 0														
Stage 1														
Stage 2														
Stage 3														
Stage 4														
Stage 5														

Stage 0: Empirical Gap Analysis (October 2025)

- Collection of Finance department audit data.
- Manual parking lot occupancy counts.
- Target: Validation of 15-20% fraud rate (Colombo Dockyard PLC, 2025).

Stage 1: POC Build (November 2025)

- Collection of 500+ training images.
- Training of YOLOv8 and CRNN models.
- Target: Detection accuracy $\geq 85\%$.

Stage 2: MVP + User Interface (December 2025 - January 2026)

- Implementation of dual-camera capture.
- Development of LED display interface.
- Implementation of fuzzy matching logic.

- Target: End-to-end latency <2 seconds.

Stage 3: Audit System Completion (February 2026)

- Implementation of comprehensive audit logging.
- Development of database integrity tools.
- Creation of export modules (CSV/JSON).

Stage 4: Testing & Validation (March 2026)

- Performance profiling across diverse conditions.
- Usability testing with security personnel.
- Target: p95 latency ≤ 1.5 seconds (Colombo Dockyard PLC, 2025).

Stage 5: Finalization (April 2026)

- Production deployment at Gate 3.
- Final documentation and thesis report.

Resource Allocation

- Hardware: Development laptop (RTX 3060), Dual 2K WDR/IR cameras, LED display panel.
- Software: Python, PyTorch, TensorRT, MS SQL Server.
- Human Resources: Developer, Supervisor, Security Personnel (testing).

Chapter 08: Risk Analysis

Technical Risks

Risk 1: Insufficient Detection Accuracy in Maritime Conditions

Likelihood: Medium | Impact: High

Salt spray and heavy rain may degrade plate visibility. *Mitigation:* The training dataset will encompass worst-case conditions. Dual-camera redundancy is implemented to provide independent viewing angles.

Risk 2: GPU Thermal Throttling

Likelihood: High | Impact: Medium

High ambient temperatures (30-35°C) may cause throttling. *Mitigation:* Active cooling fans and white enclosures will be used. If necessary, hardware will be relocated to the air-conditioned security office.

Risk 3: Motorcycle Detection Gap

Likelihood: Medium | Impact: Medium

Motorcycle plates are small and often non-standard. *Mitigation:* A specific motorcycle detection class will be trained. Fallback manual entry workflow is designed for undetected vehicles.

Risk 4: Database Performance Degradation

Likelihood: Low | Impact: Medium

Query speed may decrease with log volume. *Mitigation:* B-tree indexes and monthly archival procedures will be implemented.

Operational Risks

Risk 5: Security Personnel Resistance

Likelihood: Medium | Impact: High

Staff may bypass the system during peak hours. *Mitigation:* Security staff will be involved in early testing. The interface is designed with simple ACCEPT/REJECT controls.

Risk 6: Hazardous Area Compliance

Likelihood: Low | Impact: Critical

Gate 3 may be a classified Zone 2 area. *Mitigation:* Zone maps will be verified. Certified explosion-proof enclosures will be procured if required.

Financial Risks

Risk 7: Hardware Procurement Delays

Likelihood: Medium | Impact: Medium

Import delays may affect the timeline. *Mitigation:* Orders will be placed early. Development will proceed using recorded footage if hardware is delayed.

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